Scalable Parallel K-Means Clustering – Project Synopsis

PROJECT SYNOPSIS

Scalable Parallel K-Means Clustering for Big Data Using MPI and CUDA

Problem Statement 5 – Parallel Computer Architecture & Programming

Mini-Project Synopsis

Project Title

Scalable Parallel K-Means Clustering for Big Data Using MPI and CUDA

Team Member Details

1. Tirth Miteshkumar Patel

a. Registration number: 240905272

b. Roll No: 27

c. Section: CSE D

Role: Distributed-memory MPI design, dataset partitioning, synchronization, and scaling analysis.

2. Dispo

a. Registration Number: to be finalized

b. Roll No: to be finalized

c. Section: to be finalized

Role: CUDA/GPU-side assignment kernel, memory optimization, and performance validation for the point-to-centroid computation.

Abstract

Clustering is a fundamental unsupervised learning technique used to discover groups or patterns in large datasets. K-Means is widely used because of its conceptual simplicity and computational efficiency, but its repeated distance calculations over millions or billions of high-dimensional data points make conventional single-processor execution increasingly expensive. Problem Statement 5 identifies scalable parallel clustering as a high-performance computing problem and proposes distributing the dataset across MPI ranks while accelerating the distance/assignment phase using CUDA, followed by global centroid updates through MPI collective communication. This project therefore combines two complementary workstreams: the MPI layer manages dataset distribution and synchronization, while Dispo contributes the GPU-side CUDA implementation responsible for accelerating the assignment stage and validating performance gains.

Work allocation

- Tirth: MPI rank management, partition planning, all-reduce centroid updates, and benchmark harnesses.
- Dispo: CUDA assignment kernel, device memory layout, optimization, and GPU-side profiling/validation.